

Lesson 4: SLR Inference and Prediction

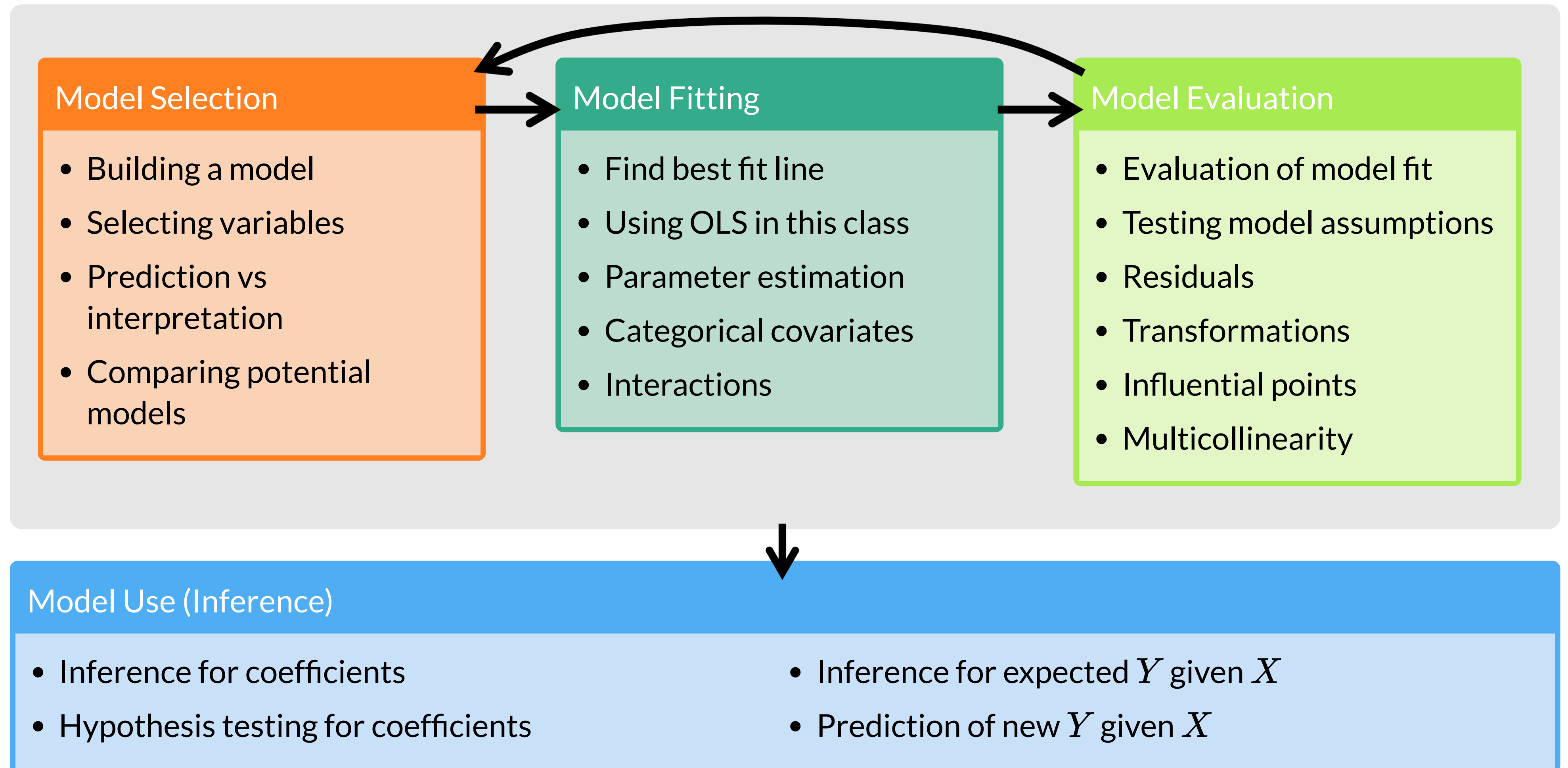
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Learning Objectives

1. Estimate the variance of the residuals
2. Using a hypothesis test, determine if there is enough evidence that population slope β_1 is not 0 (applies to β_0 as well)
3. Calculate and report the estimate and confidence interval for the population slope β_1 (applies to β_0 as well)
4. Calculate and report the estimate and confidence interval for the expected/mean response given X

Process of regression data analysis



Let's remind ourselves of the model that we fit last lesson

- We fit Gapminder data with cell phones as our independent variable and life expectancy as our dependent variable
- We used OLS to find the coefficient estimates of our best-fit line

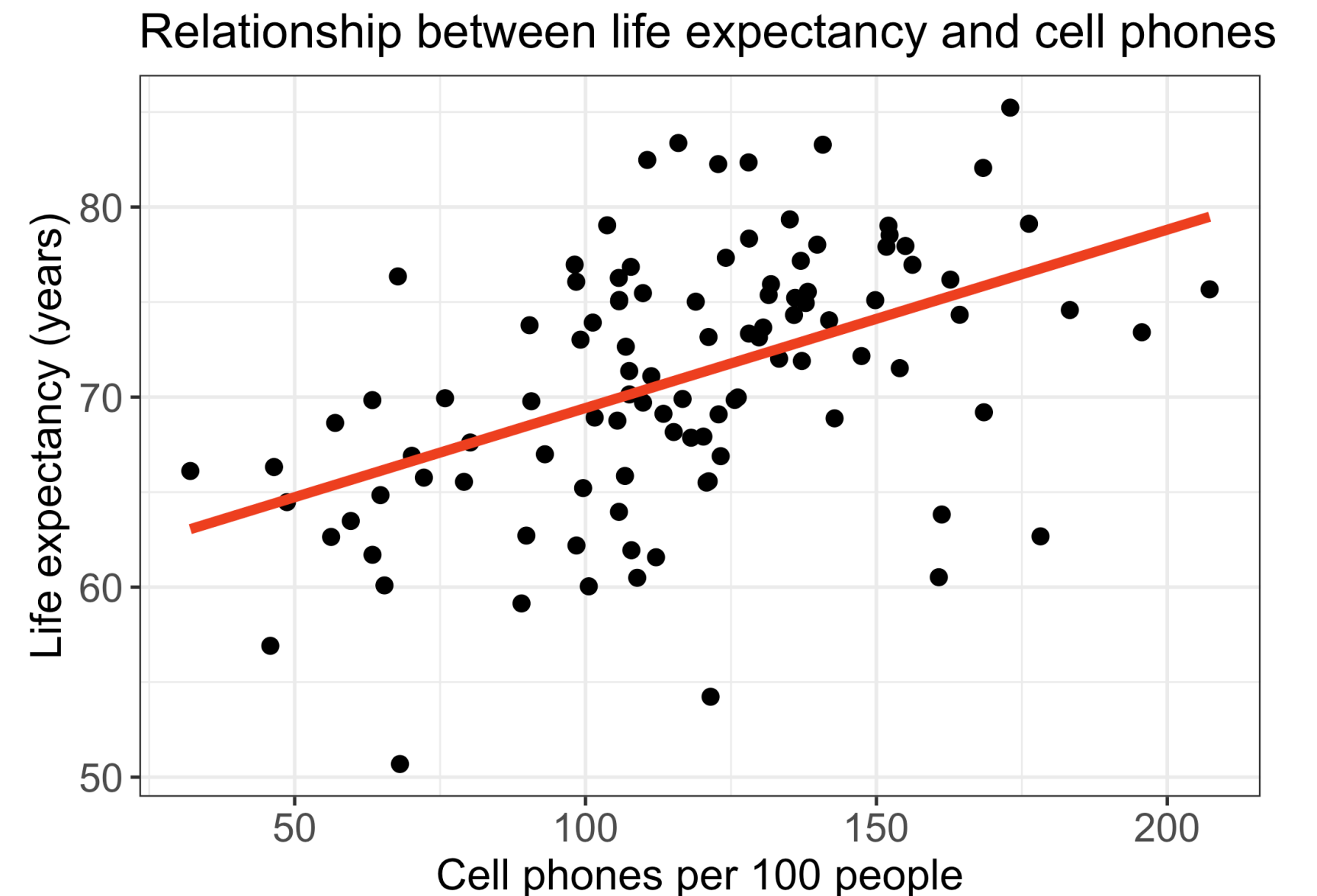
```
1 model1 <- gapm %>%  
2   lm(formula = life_exp ~ cell_phones_100)
```

► Code

term	estimate	std.error	statistic	p.value
(Intercept)	60.04	2.06	29.21	0.00
cell_phones_100	0.09	0.02	5.55	0.00

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot X$$

$$\widehat{\text{life expectancy}} = 50.9 + 0.232 \cdot \text{cell phones}$$



Fitted line is derived from the population SLR model

The (population) regression model is denoted by:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- β_0 and β_1 are **unknown** population parameters
- ϵ (epsilon) is the error about the line
 - It is assumed to be a random variable with a...
 - Normal distribution with mean 0 and constant variance σ^2
 - i.e. $\epsilon \sim N(0, \sigma^2)$

Poll Everywhere Question 1

Learning Objectives

1. Estimate the variance of the residuals

2. Using a hypothesis test, determine if there is enough evidence that population slope β_1 is not 0 (applies to β_0 as well)
3. Calculate and report the estimate and confidence interval for the population slope β_1 (applies to β_0 as well)
4. Calculate and report the estimate and confidence interval for the expected/mean response given X

$\hat{\sigma}^2$: Needed ingredient for inference

- Recall our population model residuals are distributed by $\epsilon \sim N(0, \sigma^2)$
 - And our estimated residuals are $\hat{\epsilon} \sim N(0, \hat{\sigma}^2)$
- Hence, the *variance* of the errors (residuals) is estimated by $\hat{\sigma}^2$

$$\hat{\sigma}^2 = S_{y|x}^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \frac{1}{n-2} SSE = MSE$$

$\hat{\sigma}^2$: I hope R can calculate that for me... (1/2)

- The *standard deviation* $\hat{\sigma}$ is given in the R output as the **Residual standard error**
 - 4th line from the bottom in the **summary()** output of the model:

```
1 summary(model1)
```

Call:

```
lm(formula = life_exp ~ cell_phones_100, data = .)
```

Residuals:

Min	1Q	Median	3Q	Max
-17.211	-3.268	0.615	3.818	12.449

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	60.04051	2.05567	29.207	< 2e-16 ***
cell_phones_100	0.09384	0.01692	5.546	2.27e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.964 on 103 degrees of freedom

Multiple R-squared: 0.23, Adjusted R-squared: 0.2225

F-statistic: 30.76 on 1 and 103 DF, p-value: 2.271e-07

$\hat{\sigma}^2$: I hope R can calculate that for me... (2/2)

- It can!!

```
1 m1_sum = summary(model1)
2 m1_sum$sigma
```

```
[1] 5.964089
```

```
1 # number of observations (pairs of data) used to run the model
2 n = nobs(model1)
3 n
```

```
[1] 105
```

$\hat{\sigma}^2$ to SSE

- Recall how we minimized the SSE to find our line of best fit
- SSE and $\hat{\sigma}^2$ are closely related:

$$\begin{aligned}\hat{\sigma}^2 &= \frac{1}{n-2} SSE \\ 6.142^2 &= \frac{1}{80-2} SSE \\ SSE &= 103 \cdot 6.142^2 = 2942.48\end{aligned}$$

- 2942.48 is the smallest sums of squares of all possible regression lines through the data

Learning Objectives

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3. Calculate and report the estimate and confidence interval for the population slope β_1 (applies to β_0 as well)

4. Calculate and report the estimate and confidence interval for the expected/mean response given X

Do we trust our estimate $\hat{\beta}_1$?

- So far, we have shown that we think the estimate is 0.094
- $\hat{\beta}_1$ (coefficient estimate) uses our sample data to estimate the population parameter β_1
- Inference helps us figure out *mathematically* how much we trust our best-fit line
- Are we certain that the relationship between X and Y that we estimated reflects the true, underlying relationship?

Poll Everywhere Question 2

Inference for the population slope: hypothesis test and CI

Population model

line + random “noise”

$$Y = \beta_0 + \beta_1 \cdot X + \varepsilon$$

with $\varepsilon \sim N(0, \sigma^2)$

σ^2 is the variance of the residuals

Sample best-fit (least-squares) line

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot X$$

Note: Some sources use b instead of $\hat{\beta}$

We have two options for inference:

1. Conduct the **hypothesis test**

$$H_0 : \beta_1 = 0$$

vs. $H_A : \beta_1 \neq 0$

Note: R reports p-values for 2-sided tests

2. Construct a **95% confidence interval** for the **population slope β_1**

Learning Objectives

1. Estimate the variance of the residuals

2. Using a hypothesis test, determine if there is enough evidence that population slope β_1 is not 0 (applies to β_0 as well)

3. Calculate and report the estimate and confidence interval for the population slope β_1 (applies to β_0 as well)

4. Calculate and report the estimate and confidence interval for the expected/mean response given X

Reference: Steps in a Hypothesis Test

1. Check the **assumptions**
 - What sampling distribution are you using? What assumptions are required for it?
2. Set the **level of significance** α
3. Specify the **null** (H_0) and **alternative** (H_A) **hypotheses**
 - In symbols and/or in words
 - Alternative: one- or two-sided?
4. Specify the test statistic and its **distribution under the null**
5. Calculate the **test statistic**.
6. Calculate the **p-value** based on the observed test statistic and its sampling distribution
7. Write a **conclusion** to the hypothesis test
 - Do we reject or fail to reject H_0 ?
 - Write a conclusion in the context of the problem

Steps for hypothesis test for population slope β_1 (using t-test)

1. Check the **assumptions**

2. Set the **level of significance**

- Often we use $\alpha = 0.05$

3. Specify the **null** (H_0) and **alternative** (H_A) **hypotheses**

- Often, we are curious if the coefficient is 0 or not:

$$H_0 : \beta_1 = 0$$

vs. $H_A : \beta_1 \neq 0$

4. Specify the test statistic and its **distribution under the null**

- The test statistic is t , and follows a Student's t-distribution.

5. Calculate the **test statistic**.

- The calculated **test statistic** for $\hat{\beta}_1$ is

$$t = \frac{\hat{\beta}_1 - \beta_1}{\text{SE}_{\hat{\beta}_1}} = \frac{\hat{\beta}_1}{\text{SE}_{\hat{\beta}_1}}$$

when we assume $H_0 : \beta_1 = 0$ is true.

6. Calculate the **p-value**

- We are generally calculating: $2 \cdot P(T > t)$

7. Write a **conclusion**

- We (reject/fail to reject) the null hypothesis that the slope is 0 at the $100\alpha\%$ significance level. There is (sufficient/insufficient) evidence that there is significant association between (Y) and (X) (p-value = $P(T > t)$).

Standard error of fitted slope $\hat{\beta}_1$



$$SE_{\hat{\beta}_1} = \frac{s_{\text{residuals}}}{s_x \sqrt{n - 1}}$$

$SE_{\hat{\beta}_1}$ is the **variability** of the statistic $\hat{\beta}_1$

- $s_{\text{residuals}}^2$ is the sd of the residuals
- s_x is the sample sd of the explanatory variable x
- n is the sample size, or the number of (complete) pairs of points

Calculating standard error for $\hat{\beta}_1$ (1/2)



- Option 1: Calculate using the formula

```
1 glance(model1)
# A tibble: 1 × 12
#   r.squared adj.r.squared sigma statistic    p.value    df logLik   AIC   BIC
#   <dbl>      <dbl> <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>
1   0.230      0.222  5.96      30.8 0.000000227     1 -335.  677.  685.
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>

1 # standard deviation of the residuals (Residual standard error in summary() output)
2 (s_resid <- glance(model1)$sigma)
[1] 5.964089

1 # standard deviation of x's
2 (s_x <- sd(gapm$cell_phones_100, na.rm=T))
[1] 34.56469

1 # number of pairs of complete observations
2 (n <- nobs(model1))
[1] 105

1 (se_b1 <- s_resid/(s_x * sqrt(n-1))) # compare to SE in regression output
[1] 0.01691978
```


Calculating standard error for $\hat{\beta}_1$ (2/2)



- Option 2: Use regression table

```
1 # recall model1_b1 is regression table restricted to b1 row
2 model1_b1 <-tidy(model1) %>% filter(term == "cell_phones_100")
3 model1_b1 %>% gt() %>%
4   tab_options(table.font.size = 45) %>% fmt_number(decimals = 4)
```

term	estimate	std.error	statistic	p.value
cell_phones_100	0.0938	0.0169	5.5461	0.0000

Some important notes

- Today we are discussing the hypothesis test for a **single** coefficient
- The test statistic for a single coefficient follows a Student's t-distribution
 - It can also follow an F-distribution, but we will discuss this more with multiple linear regression and multi-level categorical covariates
- Single coefficient testing can be done on any coefficient, but it is most useful for continuous covariates or binary covariates
 - This is because testing the single coefficient will still tell us something about the overall relationship between the covariate and the outcome
 - We will talk more about this with multiple linear regression and multi-level categorical covariates

Poll Everywhere Question 3

Life expectancy example: hypothesis test for population slope β_1 (1/4)

- Steps 1-4 are setting up our hypothesis test: not much change from the general steps
1. Check the **assumptions**: We have met the underlying assumptions (checked in our Model Evaluation step)
 2. Set the **level of significance**
 - Often we use $\alpha = 0.05$
 3. Specify the **null** (H_0) and **alternative** (H_A) **hypotheses**
 - We are testing if the slope is 0 or not:

$$H_0 : \beta_1 = 0$$

vs. $H_A : \beta_1 \neq 0$

4. Specify the test statistic and its **distribution under the null**
 - The test statistic is t , and follows a Student's t-distribution.

Life expectancy example: hypothesis test for population slope β_1 (2/4)

5. Calculate the **test statistic**

Option 1: Calculate the test statistic using the values in the regression table

```
1 model1_b1 <- tidy(model1) %>%
2   filter(term == "cell_phones_100")
3 model1_b1 %>% gt() %>%
4   tab_options(table.font.size = 40) %>%
5   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value
cell_phones_100	0.094	0.017	5.546	0.000

```
1 (TestStat_b1 <- model1_b1$estimate /
2   model1_b1$std.error)
```

```
[1] 5.546063
```

Option 2: Get the test statistic value (t^*) from **R**

```
1 model1_b1 %>% gt() %>%
2   tab_options(table.font.size = 40) %>%
3   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value
cell_phones_100	0.094	0.017	5.546	0.000

Life expectancy example: hypothesis test for population slope β_1 (3/4)

6. Calculate the **p-value**

- The p -value is the *probability of obtaining a test statistic just as extreme or more extreme than the **observed** test statistic assuming the null hypothesis H_0 is true*
- We know the probability distribution of the test statistic (the *null distribution*) assuming H_0 is true
- Statistical theory tells us that the test statistic t can be modeled by a t -distribution with $df = n - 2$.
 - We had 105 countries' data, so $n = 105$

Option 1: Use `pt()` and our calculated test statistic

```
1 (pv = 2*pt(TestStat_b1,  
2           df=n-2,  
3           lower.tail=F))
```

```
[1] 2.271176e-07
```

Option 2: Use the regression table output

```
1 model1_b1 %>% gt() %>%  
2   tab_options(table.font.size = 40) %>%  
3   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value
cell_phones_100	0.094	0.017	5.546	0.000

Life expectancy example: hypothesis test for population slope β_1 (4/4)

7. Write a conclusion

We reject the null hypothesis that the slope is 0 at the 5% significance level. There is sufficient evidence that there is significant association between life expectancy and number of cell phones per 100 people (p-value < 0.0001).

Note on hypothesis testing using R

- We can basically combine Step 5 and 6 if we are using the “Option 2” route
- In our assignments: if you use Option 2, Step 5 and 6 become one
 - Unless I specifically ask for the test statistic!!

Life expectancy ex: hypothesis test for population intercept β_0 (1/4)

1. Check the **assumptions**: We have met the underlying assumptions (checked in our Model Evaluation step)
2. Set the **level of significance**
 - Often we use $\alpha = 0.05$
3. Specify the **null** (H_0) and **alternative** (H_A) **hypotheses**
 - We are testing if the intercept is 0 or not:

$$H_0 : \beta_0 = 0$$

vs. $H_A : \beta_0 \neq 0$

4. Specify the test statistic and its **distribution under the null**
 - This is the same as the slope. The test statistic is t , and follows a Student's t-distribution.

Life expectancy ex: hypothesis test for population intercept β_0 (2/4)

5. Calculate the **test statistic**

Option 1: Calculate the test statistic using the values in the regression table

```
1 model1_b0 <- tidy(model1) %>%  
2   filter(term == "(Intercept)")  
3 model1_b0 %>% gt() %>%  
4   tab_options(table.font.size = 40) %>%  
5   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value
(Intercept)	60.041	2.056	29.207	0.000

```
1 (TestStat_b0 <- model1_b0$estimate /  
2   model1_b0$std.error)
```

```
[1] 29.20728
```

Option 2: Get the test statistic value (t^*) from **R**

```
1 model1_b0 %>% gt() %>%  
2   tab_options(table.font.size = 40) %>%  
3   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value
(Intercept)	60.041	2.056	29.207	0.000

Life expectancy ex: hypothesis test for population intercept β_0 (3/4)

6. Calculate the **p-value**

Option 1: Use `pt()` and our calculated test statistic

```
1 (pv = 2*pt(TestStat_b0,  
2           df=n-2,  
3           lower.tail=F))
```

```
[1] 1.215444e-51
```

Option 2: Use the regression table output

```
1 model1_b0 %>% gt() %>%  
2   tab_options(table.font.size = 40) %>%  
3   fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value
(Intercept)	60.041	2.056	29.207	0.000

Life expectancy ex: hypothesis test for population intercept β_0 (4/4)

7. Write a conclusion

We reject the null hypothesis that the intercept is 0 at the 5% significance level. There is sufficient evidence that the intercept for the association between life expectancy and number of cell phones per 100 people is different from 0 (p-value < 0.0001).

- Note: if we fail to reject H_0 , then we *could* decide to remove the intercept from the model to force the regression line to go through the origin (0,0) if it makes sense to do so for the application
 - Not typical to remove the intercept

Learning Objectives

1. Estimate the variance of the residuals
2. Using a hypothesis test, determine if there is enough evidence that population slope β_1 is not 0 (applies to β_0 as well)
3. Calculate and report the estimate and confidence interval for the population slope β_1 (applies to β_0 as well)
4. Calculate and report the estimate and confidence interval for the expected/mean response given X

Inference for the population slope: hypothesis test and CI

Population model

line + random “noise”

$$Y = \beta_0 + \beta_1 \cdot X + \varepsilon$$

with $\varepsilon \sim N(0, \sigma^2)$

σ^2 is the variance of the residuals

Sample best-fit (least-squares) line

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot X$$

Note: Some sources use b instead of $\hat{\beta}$

We have two options for inference:

1. Conduct the **hypothesis test**

$$H_0 : \beta_1 = 0$$

vs. $H_A : \beta_1 \neq 0$

Note: R reports p-values for 2-sided tests

2. Construct a **95% confidence interval** for the **population slope β_1**

Confidence interval for population slope β_1

Recall the general CI formula:

$$\hat{\beta}_1 \pm t_{\alpha, n-2}^* \cdot SE_{\hat{\beta}_1}$$

To construct the confidence interval, we need to:

- Set our α -level
- Find $\hat{\beta}_1$
- Calculate the t_{n-2}^*
- Calculate $SE_{\hat{\beta}_1}$

Calculate CI for population slope β_1 (1/2)

$$\hat{\beta}_1 \pm t^* \cdot SE_{\beta_1}$$

where t^* is the t -distribution critical value with $df = n - 2$.

- **Option 1:** Calculate using each value

Save values needed for CI:

```
1 b1 <- model1_b1$estimate
2 SE_b1 <- model1_b1$std.error
```

```
1 nobs(model1) # sample size n
```

```
[1] 105
```

```
1 (tstar <- qt(.975, df = n - 2))
```

```
[1] 1.983264
```

Use formula to calculate each bound

```
1 (CI_LB <- b1 - tstar*SE_b1)
```

```
[1] 0.06028178
```

```
1 (CI_UB <- b1 + tstar*SE_b1)
```

```
[1] 0.1273946
```


Calculate CI for population slope β_1 (2/2)

$$\hat{\beta}_1 \pm t^* \cdot SE_{\beta_1}$$

where t^* is the t -distribution critical value with $df = n - 2$.

- Option 2: Use the regression table

```
1 tidy(model1, conf.int = T) %>% gt() %>%
2   tab_options(table.font.size = 40) %>% fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	60.041	2.056	29.207	0.000	55.964	64.117
cell_phones_100	0.094	0.017	5.546	0.000	0.060	0.127

Interpreting the coefficient estimate of the population slope with CIs

- When we report our results to someone else, we don't usually show them our full hypothesis test
 - In an informal setting, someone may want to see it
- Typically, we report the estimate with the confidence interval
 - From the confidence interval, your audience can also deduce the results of a hypothesis test
- Once we found our CI, we often just write the interpretation of the coefficient estimate:

General statement for population slope inference

For every increase of 1 unit in the X -variable, there is an expected/average (pick one) increase of $\hat{\beta}_1$ units in the Y -variable (95%: LB, UB).

- **In our example:** For every additional cell phone per 100 people, life expectancy increases, on average, 0.094 years (95% CI: 0.06, 0.127)

Usually three options for your interpretations

- **Option 1:** For every additional cell phone per 100 people, life expectancy increases, **on average**, 0.094 years (95% CI: 0.06, 0.127)
- **Option 2:** For every additional cell phone per 100 people, **average** life expectancy increases 0.094 years (95% CI: 0.06, 0.127)
- **Option 3:** For every additional cell phone per 100 people, **expected** life expectancy increases 0.094 years (95% CI: 0.06, 0.127)

Poll Everywhere Question 4

For reference: quick CI for β_0

- Calculate CI for population **intercept** β_0 : $\hat{\beta}_0 \pm t^* \cdot SE_{\beta_0}$

where t^* is the t -distribution critical value with $df = n - 2$

- Use the regression table

```
1 tidy(model1, conf.int = T) %>% gt() %>%  
2   tab_options(table.font.size = 40) %>% fmt_number(decimals = 3)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	60.041	2.056	29.207	0.000	55.964	64.117
cell_phones_100	0.094	0.017	5.546	0.000	0.060	0.127

General statement for population intercept inference

The expected outcome for the Y -variable is $(\hat{\beta}_0)$ when the X -variable is 0 (95% CI: LB, UB).

- For example:** The average life expectancy is 60.04 years when the number of cell phones per person is 0 (95% CI: 55.96, 64.12).

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4. Calculate and report the estimate and confidence interval for the expected/mean response given X

Finding a mean response given a value of our independent variable

term	estimate	std.error	statistic	p.value
(Intercept)	60.041	2.056	29.207	0.000
cell_phones_100	0.094	0.017	5.546	0.000

$$\widehat{\text{life expectancy}} = 60.041 + 0.094 \cdot \text{cell phones}$$

- What is the expected/predicted life expectancy for any country with 60 cell phones per 100 people?

$$\widehat{\text{life expectancy}} = 60.041 + 0.094 \cdot 60 = 65.671$$

```
1 (y_60 <- tidy(model1)$estimate[1] + tidy(model1)$estimate[2]*60)
```

```
[1] 65.6708
```

- How do we interpret the expected value?
 - We sometimes call this “predicted” value, since we can technically use a number (of cell phones) that is not in our sample
- How variable is it?

Mean response/prediction with regression line

Recall the population model:

line + random “noise”

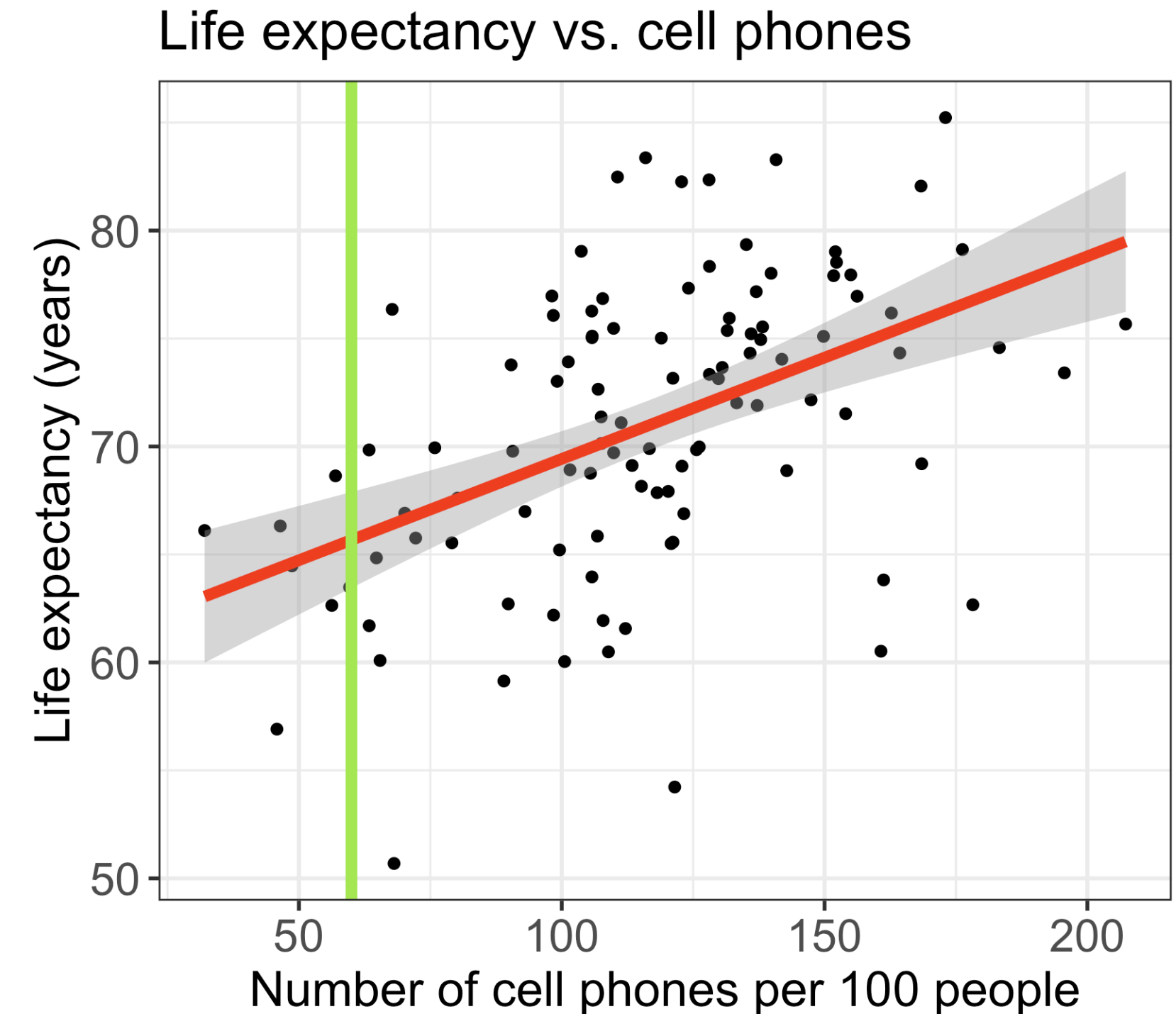
$$Y = \beta_0 + \beta_1 \cdot X + \varepsilon$$

with $\varepsilon \sim N(0, \sigma^2)$

- When we take the expected value, at a given value X^* , the average expected response at X^* is:

$$\hat{E}[Y|X^*] = \hat{\beta}_0 + \hat{\beta}_1 X^*$$

- These are the points on the regression line
- The mean responses have variability, and we can calculate a CI for it, for every value of X^*



CI for population mean response ($\hat{E}[Y|X^*]$ or $\mu_{Y|X^*}$)

$$\hat{E}[Y|X^*] \pm t_{n-2}^* \cdot SE_{\hat{E}[Y|X^*]}$$

$$SE_{\hat{E}[Y|X^*]} = s_{\text{residuals}} \sqrt{\frac{1}{n} + \frac{(X^* - \bar{X})^2}{(n-1)s_X^2}}$$

- $\hat{E}[Y|X^*]$ is the predicted value at the specified point X^* of the explanatory variable
- $s_{\text{residuals}}^2$ is the sd of the residuals
- n is the sample size, or the number of (complete) pairs of points
- $\bar{X}$ is the sample mean of the explanatory variable x
- s_X is the sample sd of the explanatory variable X
- Recall that t_{n-2}^* is calculated using `qt()` and depends on the confidence level $(1 - \alpha)$

Example Option 1: CI for mean response $\mu_{Y|X^*}$

Find the 95% CI's for mean life expectancy for 60 cell phones per 100 people:

$$\begin{aligned} & \hat{E}[Y|X^*] \pm t_{n-2}^* \cdot SE_{\hat{E}[Y|X^*]} \\ & 65.671 \pm 1.983 \cdot s_{residuals} \sqrt{\frac{1}{n} + \frac{(X^* - \bar{x})^2}{(n-1)s_x^2}} \\ & 65.671 \pm 1.983 \cdot 5.964 \sqrt{\frac{1}{105} + \frac{(60 - 116.523)^2}{(105-1)34.565^2}} \\ & 65.671 \pm 1.983 \cdot 1.12 \\ & 65.671 \pm 2.22 \\ & (63.45, 67.891) \end{aligned}$$

Example Option 2: CI for mean response $\mu_{Y|X^*}$

Find the 95% CI's for mean life expectancy for 60 and 80 cell phones per 100 people, respectively.

- Use the base R `predict()` function
- Requires specification of a `newdata` "value"
 - The `newdata` value is X^*
 - This has to be in the format of a data frame though
 - with column name identical to the predictor variable in the model

```
1 newdata <- data.frame(cell_phones_100 = c(60, 80))
2 newdata
```

```
  cell_phones_100
1              60
2              80
```

```
1 predict(model1,
2         newdata=newdata,
3         interval="confidence")
```

```
      fit      lwr      upr
1 65.67080 63.45044 67.89117
2 67.54757 65.86395 69.23118
```

Interpretation

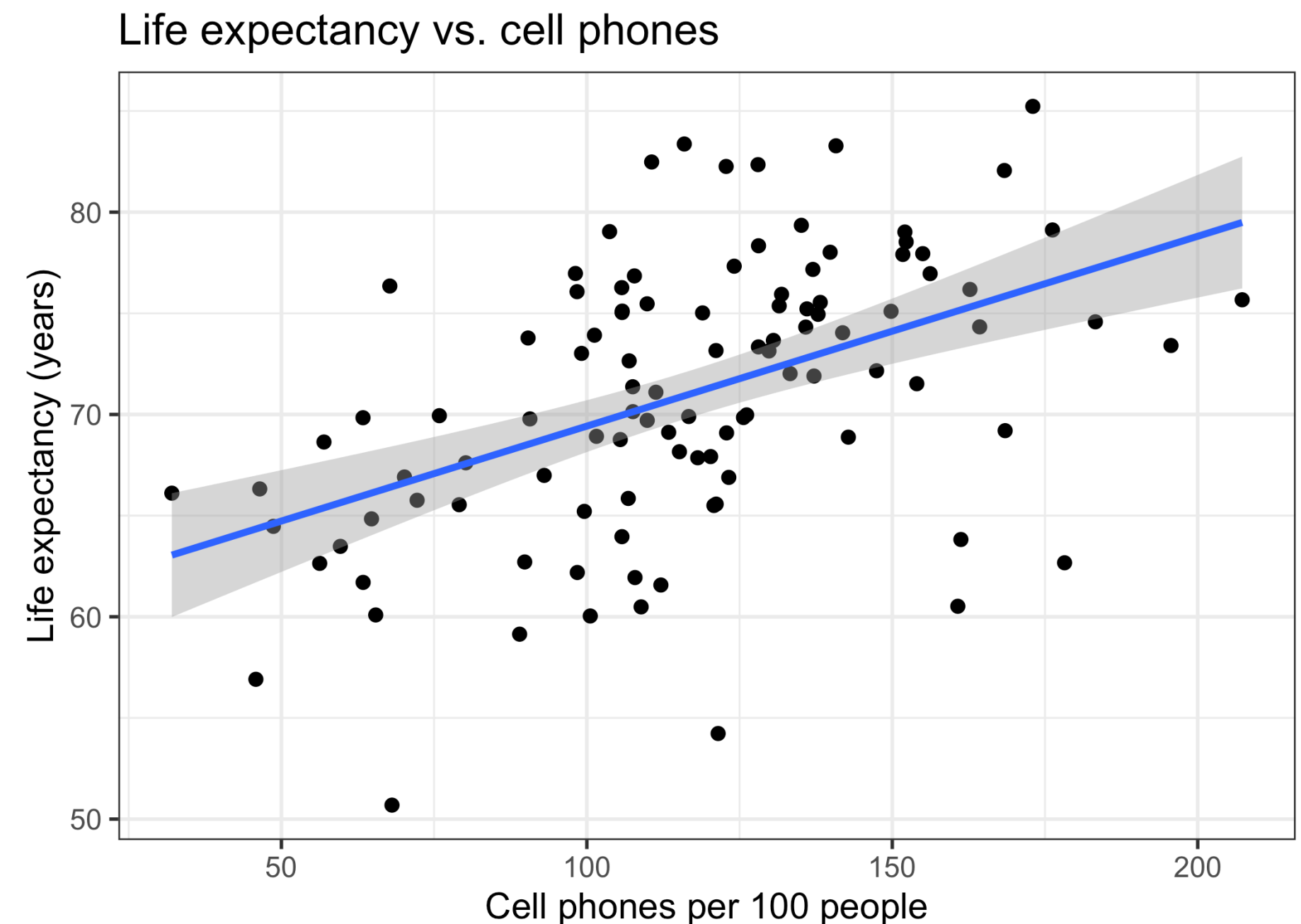
We are 95% confident that the **average** life expectancy for a country with a 60 cell phones per 100 people will be between 63.45 and 67.89 years.

Poll Everywhere Question 5

Confidence bands for mean response $\mu_{Y|X^*}$

- Often we plot the CI for many values of X, creating **confidence bands**
- The confidence bands are what ggplot creates when we set `se = TRUE` within `geom_smooth`
- Think about it: for what values of X are the confidence bands (intervals) narrowest?

```
1 gapm %>%
2   ggplot(
3     aes(
4       x=cell_phones_100,
5       y=life_exp
6     )
7   ) +
8   geom_point() +
9   geom_smooth(method = lm, se=TRUE) +
10  labs(
11    x = "Cell phones per 100 people",
12    y = "Life expectancy (years)",
13    title = "Life expectancy vs. cell phone
14  )
```

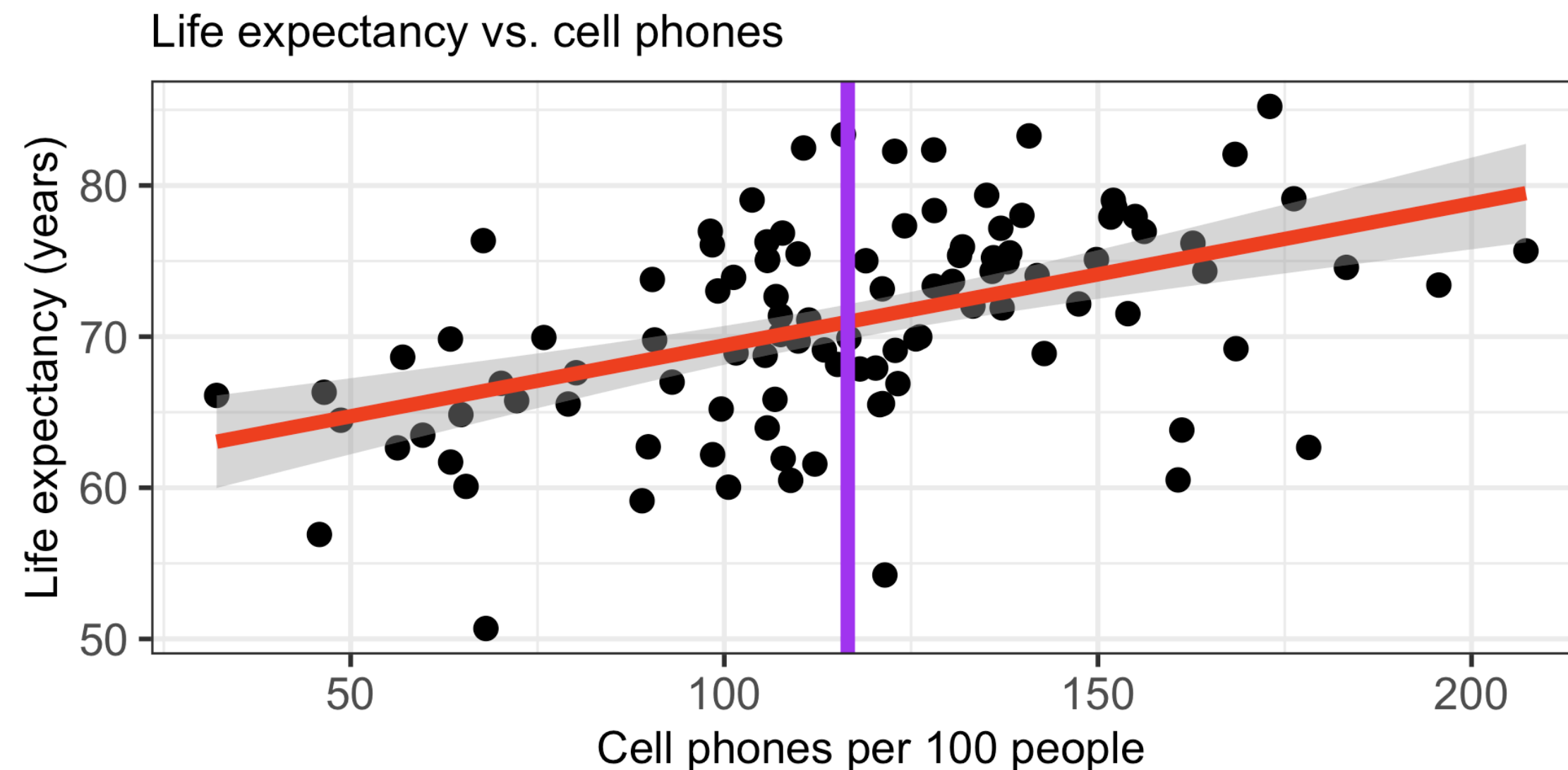


Width of confidence bands for mean response $\mu_{Y|X^*}$

- For what values of X^* are the confidence bands (intervals) narrowest? widest?

$$\hat{E}[Y|X^*] \pm t_{n-2}^* \cdot SE_{\hat{E}[Y|X^*]}$$

$$\hat{E}[Y|X^*] \pm t_{n-2}^* \cdot s_{residuals} \sqrt{\frac{1}{n} + \frac{(X^* - \bar{x})^2}{(n-1)s_x^2}}$$



Textbook readings

- Introduction to Regression Methods for Public Health Using R
 - 4.5 Interpreting p-values
 - 4.6 Predictions from the model
 - 4.7 Confidence intervals and prediction intervals
- A Progressive Introduction to Linear Models
 - Not really any good sections